

BLOOD BANK RARE DONAR NETWORK

Problem statement:

Blood banks operate in isolation, making it difficult to quickly locate rare blood types during emergencies. The lack of centralized inventory tracking leads to delays, inefficient donor engagement, and wastage of blood due to expiry. There is a need for a smart, connected system that ensures real-time availability, efficient donor matching, and secure handling of sensitive donor information.

1. Problem Understanding

“Blood banks currently work in isolation. When rare blood types like O-negative are needed, hospitals waste critical time searching manually. At the same time, blood units expire due to poor tracking and lack of coordination.”

2. Idea Clarity

“We propose a Smart Blood Inventory & Rare Donor Network that connects multiple hospitals, tracks blood units with expiry (FIFO), and instantly matches eligible donors during emergencies while ensuring data privacy.”

3. Feasibility

“This solution is feasible using simple web technologies like React and Node.js with a lightweight database. Core features like inventory tracking, donor matching, and alerts can be implemented within the hackathon timeframe.”

4. Planning

“Our plan is to first build the core modules: inventory system, donor registry, and urgent request feature. Then we will integrate them into a dashboard and demonstrate a complete flow from blood search to donor notification.”